

Causal Inference and ML: Beyond Correlation in Sports Data

Estimating Heterogeneous Effects on Marathon Data

Kristian Boroz

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Github Repository:

`github.com/kboroz/Causal_Inference_and_ML/Week07_Project_Session`

Core Research Questions

1. Prediction

Primary Objective: Identify the core drivers of **Actual Finish Time**.

Sub-Points:

- Evaluate non-linear interaction effects.
- Benchmark Linear vs. Tree Ensembles
- Identify Causal Relationships

2. Risk Optimization

Primary Objective: Isolate training triggers for **Injury Counts**.

Sub-Points:

- Model count metrics via Poisson/NB approaches.
- Locate systemic thresholds in mileage and missing rest days.

3. Causal Inference

Primary Objective: Establish if mileage *causes* speed.

Sub-Points:

- Control for selection bias and baseline fitness (*Age, VO2 Max, etc*).

Table 1: Summary Statistics of Runner Dataset

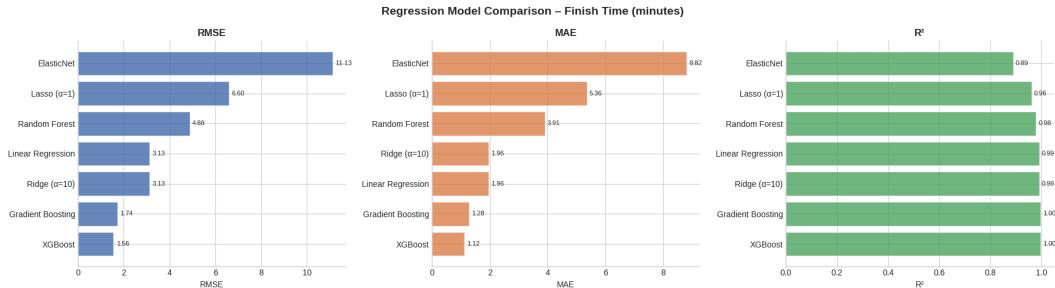
Metric	count	unique	top	freq	mean	std	min	25%	50%	75%	max
runner_id	80000	80000	R057981	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
age	80000.0	NaN	NaN	NaN	41.538412	13.848164	18.0	30.0	42.0	54.0	65.0
gender	80000	3	Female	38577	NaN	NaN	NaN	NaN	NaN	NaN	NaN
running_experience_months	80000.0	NaN	NaN	NaN	36.147413	35.453932	6.0	10.0	25.0	49.0	240.0
previous_marathon_count	80000.0	NaN	NaN	NaN	2.0047	1.419094	0.0	1.0	2.0	3.0	11.0
training_program	80000	3	Beginner	39983	NaN	NaN	NaN	NaN	NaN	NaN	NaN
motivation_level	80000.0	NaN	NaN	NaN	5.505575	2.871288	1.0	3.0	6.0	8.0	10.0
personal_best_minutes	69255.0	NaN	NaN	NaN	259.32158	33.778153	132.0	236.0	261.0	283.0	377.0
weekly_mileage_km	80000.0	NaN	NaN	NaN	29.940183	17.392518	20.0	20.0	20.6	34.3	119.994348
weekly_mileage_miles	80000.0	NaN	NaN	NaN	17.225917	7.07249	12.4	12.4	12.4	20.5	60.2
runs_per_week	80000.0	NaN	NaN	NaN	4.356975	0.896649	3.0	4.0	4.0	5.0	7.0
long_run_distance_km	80000.0	NaN	NaN	NaN	15.828567	2.358518	15.0	15.0	15.0	15.0	41.2
speed_work_sessions_per_week	80000.0	NaN	NaN	NaN	0.978862	0.645074	0.0	1.0	1.0	1.0	3.0
rest_days_per_week	80000.0	NaN	NaN	NaN	2.470087	0.69113	0.0	2.0	3.0	3.0	3.0
training_adherence_pct	80000.0	NaN	NaN	NaN	80.386075	11.804666	40.0	73.0	81.0	89.0	100.0
consecutive_weeks_no_miss	80000.0	NaN	NaN	NaN	12.326562	2.499604	2.0	11.0	12.0	14.0	16.0
cross_training_hours_per_week	72017.0	NaN	NaN	NaN	1.480117	2.364641	0.0	0.0	0.1	2.1	10.0
resting_heart_rate_bpm	80000.0	NaN	NaN	NaN	75.223213	6.483532	45.0	72.0	78.0	80.0	80.0
vo2_max	70389.0	NaN	NaN	NaN	53.757363	8.507076	35.0	47.84734	53.557431	59.45122	75.0
bmi	80000.0	NaN	NaN	NaN	22.781974	2.523397	18.0	21.0	22.7	24.5	32.0
recovery_score	80000.0	NaN	NaN	NaN	6.674585	2.067715	1.0	5.3	6.8	8.2	10.0
sleep_hours_avg	76058.0	NaN	NaN	NaN	6.965946	0.996125	5.0	6.3	7.0	7.7	10.0
injury_count	80000.0	NaN	NaN	NaN	0.551813	0.773804	0.0	0.0	0.0	1.0	5.0
injury_severity	32650	3	Minor	16958	NaN	NaN	NaN	NaN	NaN	NaN	NaN
nutrition_score	73598.0	NaN	NaN	NaN	5.604772	2.206328	1.0	4.0	6.0	7.0	10.0
hydration_consistency	73598.0	NaN	NaN	NaN	5.761964	2.105822	1.0	4.0	6.0	7.0	10.0

Data Structure - Visualization

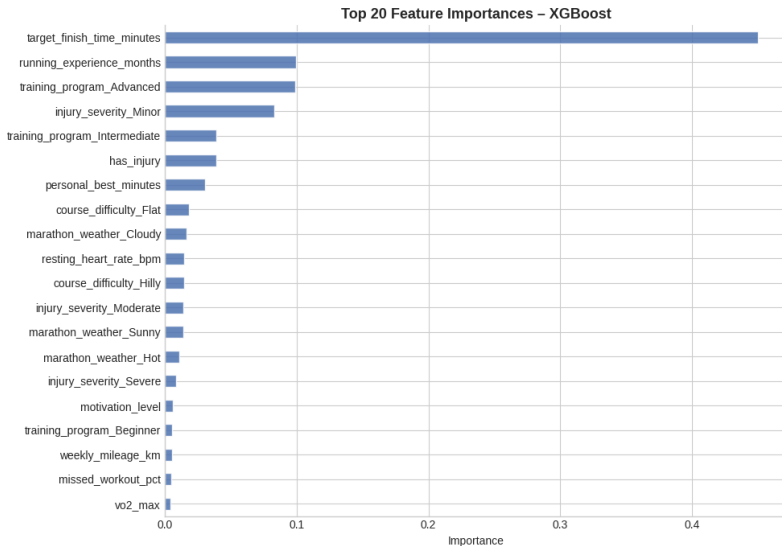
MARATHON DATASET - EDA OVERVIEW - 80,000 Runners - 40 Variables



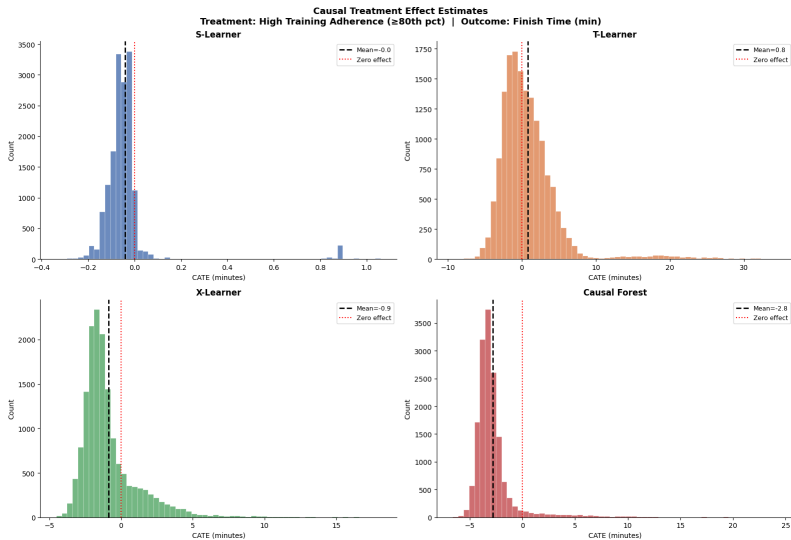
Regression Models - Results



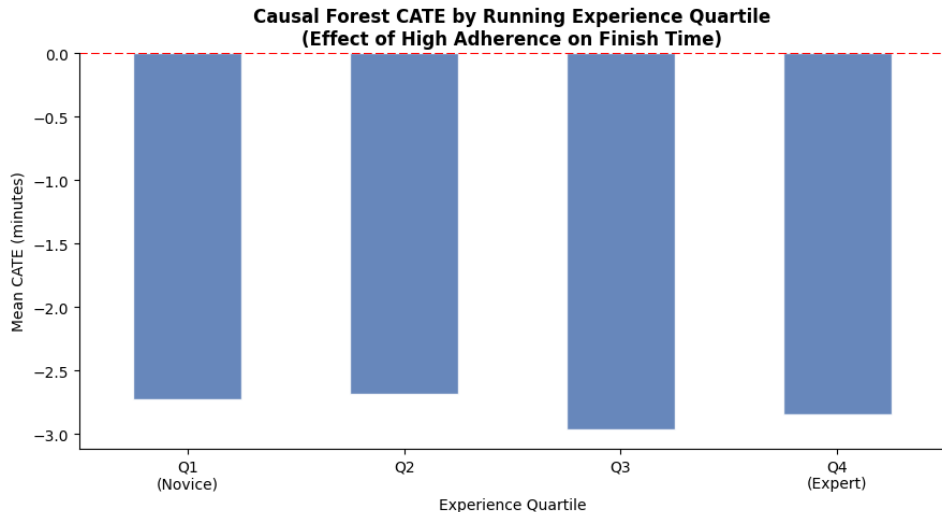
Regression Models - Feature Importance



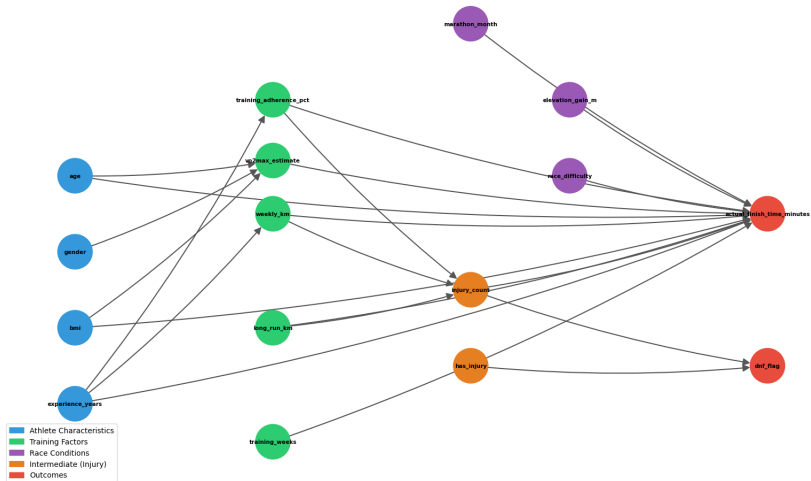
T,S,X-Learners and Causal Forest



T,S,X-Learners and Causal Forest

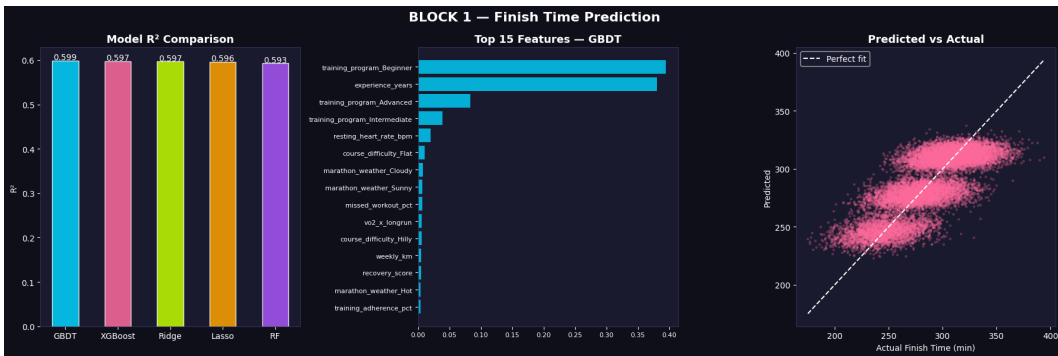


Causal DAG - Marathon Performance



Research Question - Estimation Results

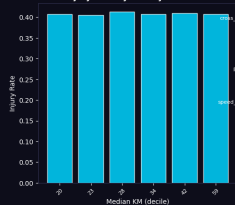
Research Question - Finishing Times



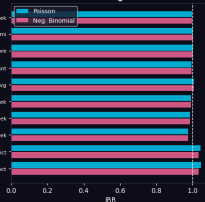
Research Question - Injuries

BLOCK 2 — Injury Risk Analysis

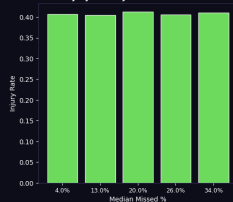
Injury Rate by Weekly KM Decile



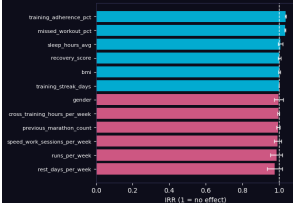
IRR: Poisson vs Negative Binomial



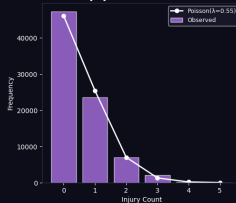
Injury Rate by Missed Workout %



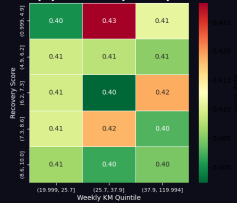
NB Model — IRR Forest Plot



Injury Count vs Poisson Fit



Injury Rate: Recovery x Weekly KM



Research Question - Speed vs. Mileage

